

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2026 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: 25AF1293PC401 - POWER SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which power plant has the lowest operating cost?
A) Thermal. B) Nuclear
C) Hydroelectric. D) Diesel
- 2 Economizer in a thermal plant is used to:
A) Heat air. B) Heat feed water
C) Cool steam. D) Remove ash
- 3 Moderator in a nuclear reactor is used to:
A) Increase neutron speed
B) Decrease neutron speed
C) Cool reactor
D) Absorb radiation
- 4 Kaplan turbine is suitable for:
A) High head B) Medium head
C) Low head. D) Very high head
- 5 The conductor commonly used in overhead transmission lines is:
A) Copper. B) ACSR. C) Gold. D) Nichrome
- 6 String efficiency is maximum when:
A) Disc voltages are unequal
B) Disc voltages are equal
C) Bottom disc voltage is maximum
D) Top disc voltage is minimum
- 7 The capacitance of a transmission line increases with:
A) Conductor spacing. B) Conductor diameter
C) Line length reduction. D) Resistance
- 8 Which insulator is used at dead ends?
A) Pin insulator. B) Suspension insulator
C) Shackle insulator. D) Strain insulator

- 9 Suspension insulators are used for:
 A) Low voltage lines. B) High voltage lines
 C) House wiring. D) Underground cables
- 10 Corona can be reduced by:
 A) Increasing conductor diameter
 B) Decreasing conductor diameter
 C) Reducing spacing
 D) Increasing resistance
- 11 Which parameter calculation is significantly affected by the presence of the ground plane underneath a transmission line?
 A) Series resistance
 B) Shunt capacitance
 C) Series inductance
 D) Surge impedance
- 12 Skin effect in a conductor causes the current to flow predominantly:
 A) Through the geometric core center of the conductor
 B) Equally throughout the entire cross-section
 C) Near the outer surface of the conductor
 D) In an axial spiral trajectory

Q2. Solve the following.

- A) Explain the schematic layout of a modern Thermal Power Station. Detail the roles of the Superheater, Economizer, and Air Preheater 6
- B) Draw the comprehensive neat layout of a Hydroelectric Power Plant. Differentiate clearly between the functions of a surge tank and a spillway 6

Q3. Solve the following.

- A) Derive the analytical mathematical expression for conductor sag when supports are located at equal horizontal levels. 6
- B) Define 'String Efficiency' for a suspension insulator assembly. Derive the mathematical expression for voltage distribution across a 3-disc insulator string. 6

Q4. Solve Any Two of the following. <https://www.batuonline.com>

- A) Derive the expression for the loop inductance of a single-phase, two-wire transmission line. Clearly account for both internal and external flux linkages. 6
- B) Explain the concept of Self-GMD (GMR) and Mutual-GMD. How do these concepts simplify the inductance calculations for asymmetrical, multi-conductor three-phase transmission networks? 6
- C) Derive the expression for the inductance per phase of a 3-phase transmission line with symmetrical spacing. 6

Q5. Solve Any Two of the following.

- A) Classify transmission lines into short, medium, and long models based on length and operational voltage. 6
- B) Derive the expressions for the sending-end voltage and current of a medium transmission line using the Nominal-T representation. Draw its complete phasor diagram. 6
- C) Define Corona Discharge and differentiate between visual and critical disruptive voltages. 6

Q6. Solve Any Two of the following.

- A) Compare radial, ring main, and interconnected distribution systems. Evaluate them based on reliability, voltage regulation, and initial capital cost. 6
- B) Define a Feeder, a Distributor, and Service Mains. Explain why a feeder is designed based on its current-carrying capacity 6
- C) Explain how AC distribution calculations differ from DC distribution calculations. Discuss the significance of calculating voltage drops with respect to a reference receiving-end voltage 6

End